

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks available |          |          |           | Maths    | Prac     |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2      | AO3      | Total     |          |          |
| 2        | (a) |  | <p>The most stable nuclei (or reference to elements near to peak i.e. Fe, Ni Ca) are therefore where the curve (or binding energy per nucleon) reaches its maximum. (1)</p> <p>Nuclei of small atomic mass number (lhs of graph) can combine to produce species of larger atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fusion. (1)</p> <p>Nuclei of large atomic mass number (on rhs of graph) break down to produce species of smaller atomic mass number, [hence larger binding energy per nucleon. Energy is released]. Fission. (1)</p> <p>Reference anywhere to there being energy released, when a <u>reduction in mass</u> occurs i.e. mass converted to energy. (1)</p> | 4               |          |          | 4         |          |          |
|          | (b) |  | <p>Mass defect = <math>4(1.00728) + 2(0.00055)</math> (1)</p> <p><math>- 4.00151 = [0.02871 \text{ u}]</math> (1)</p> <p>Energy: <math>0.02871 \times 931 = 26.7 [\text{MeV}]</math> (1)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 3        |          | 3         | 3        |          |
|          | (c) |  | <p><b>Benefit:</b> routine supply of energy (i.e. does not depend on weather) or jobs or no CO<sub>2</sub> emission (1)</p> <p><b>Issue:</b> needs secure storage of radioactive waste (products) over a extended time period or reference to long half-lives or long build time or building extra transmission power lines (1)</p> <p><b>Reasoned conclusion</b> (1)</p>                                                                                                                                                                                                                                                                                                                                             |                 |          | 3        | 3         |          |          |
|          |     |  | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>4</b>        | <b>3</b> | <b>3</b> | <b>10</b> | <b>3</b> | <b>0</b> |